Chapter 8

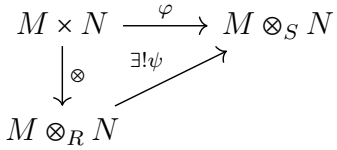
Exercise 3.2. If $f : R \to S$ is a ring homomorphism and M, N are S-modules (hence R-modules by restriction of scalars), prove there is a canonical homomorphism of R-modules $M \otimes_R N \to M \otimes_S N$.

Step where $\varphi(rm, n) = \varphi(f(r)m, n)$ seems wrong.

Proof. Let $\varphi : M \times N \to M \otimes_S N$ be defined by $\varphi(m, n) = m \otimes_S n$. φ is an R-bilinear map:

$$\varphi(rm, n) = \varphi(f(r)m, n) = f(r)m \otimes_S n = m \otimes_S f(r)n = \varphi(m, f(r)n) = \varphi(m, rn)$$

Therefore there is a unique ψ making the following diagram commute; ψ is the required canonical homomorphism between $M \otimes_R N$ and $M \otimes_S N$.



□